Here the

$$\Delta ng = n_p - n_r = 2 - 2 = 0$$

Hence Kp = Kc

$$K_P = \frac{4x^2}{(a-x)(b-x)}$$

Solved Problem

One mole of H_2 and one mole of I_2 are allowed to attain equilibrium in 1 lit container. If the equilibrium mixture contains 0.4 mole of HI. Calculate the equilibrium constant.

Given data:

$[H_2] = 1 \text{ mol } L^{-1}$ $[I_2] = 1 \text{ mol } L^{-1}$

At equilibrium, $[HI] = 0.4 \text{ mol } L^{-1}$ Kc= ?

Solution:

$$H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$$

	H_2	I_2	HI
Initial number of moles	1	1	–
Number of moles at equilibrium	1-x	1-x	2x = 0.4. x=0.2
	0.8	0.8	0.4

$$\therefore K_C = \frac{[HI]^2}{[H_2][I_2]} = \frac{0.4 \times 0.4}{0.8 \times 0.8} = 0.25$$

Dissociation of PCl_5:

Consider that 'a' moles of PCl_5 is taken in a container of volume V. Let 'x' moles of PCl_5 be dissociated into x moles of PCl_3 and x moles of Cl_2.

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

	PCl_5	PCl_3	Cl_2
Initial number of moles	a	0	0
number of moles dissociated	x	0	0
Number of moles at equilibrium	a-x	x	x
Active mass or molar concentration at equilibrium	$\frac{a-x}{V}$	$\frac{x}{V}$	$\frac{x}{V}$

Applying law of mass action,

$$K_C = \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{\left(\frac{x}{V}\right)\left(\frac{x}{V}\right)}{\left(\frac{a-x}{V}\right)} = \frac{x^2}{(a-x)V}$$

The equilibrium constant K_p can also be calculated as follows:

We know the relationship between the Kc and Kp

$$K_P = K_C (RT)^{\Delta n_g}$$

Here the

$$\Delta n_g = n_p - n_r = 2 - 1 = 1$$

Hence Kp = Kc (RT)

We know that PV = nRT

$$RT = \frac{PV}{n}$$

Where n is the total number of moles at equilibrium.

n = (a-x) + x + x = (a+x)

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